

⚠️ **EARLY-VERSION TEST MOCK-UP — ILLUSTRATES THE IDEA AND THE GOAL ONLY · NOT A DESIGN · NOT FOR CONSTRUCTION · EVERYTHING SUBJECT TO COMPLETE CHANGE**



*VERY EARLY CONCEPT ILLUSTRATION — NOT A DESIGN DOCUMENT. The SELENE design map, companion to the BRONTE Chimborazo sheet (KER-BRO-TRK-001). Two machines, one ridge: **EOS-1** (Ø1.2 m pod — built first, cargo return to Earth) and **EOS-2** (Ø2.0 m pod — second machine, second pad, bigger cargo, future). The site map is rendered from real NASA LRO/LOLA laser topography (20 m/px polar DEM); only the track alignment drawn on it is conceptual. All physics figures re-derived June 12, 2026.*

ANNEX B

SELENE — South Pole Design Map

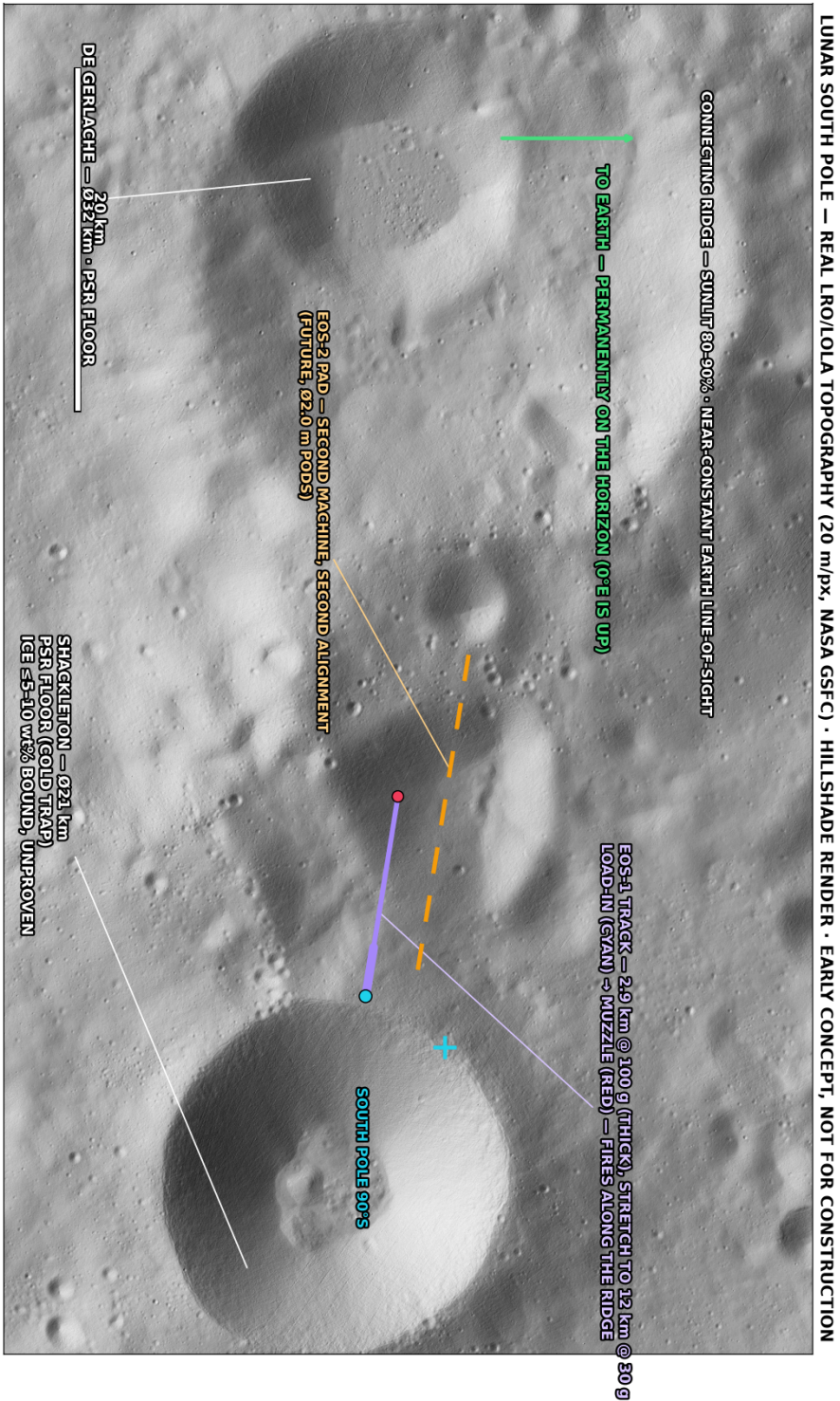
KER-SEL-TRK-002 · REV D · EARLY-VERSION TEST MOCK-UP · NOT FOR CONSTRUCTION

WHAT IS INSIDE

- ▶ Panel A — Site: real LRO/LOLA laser topography, Shackleton / Connecting Ridge ~89.7°S
- ▶ Panel A2 — Where on the Moon · what does not exist here · two machines (EOS-1 / EOS-2)
- ▶ Panel B — EOS-1 open-track section: no tube, the Moon is the vacuum vessel
- ▶ Panel C — Side profile: the raised track on its regolith embankment (true 1:1)
- ▶ Charts — track-length vs g, Δv destinations, ridge illumination, engineering

Each large panel is on its own page, rotated to read at full size — turn the page (or your screen) a quarter-turn clockwise.

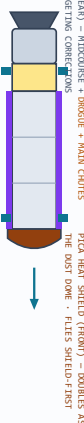
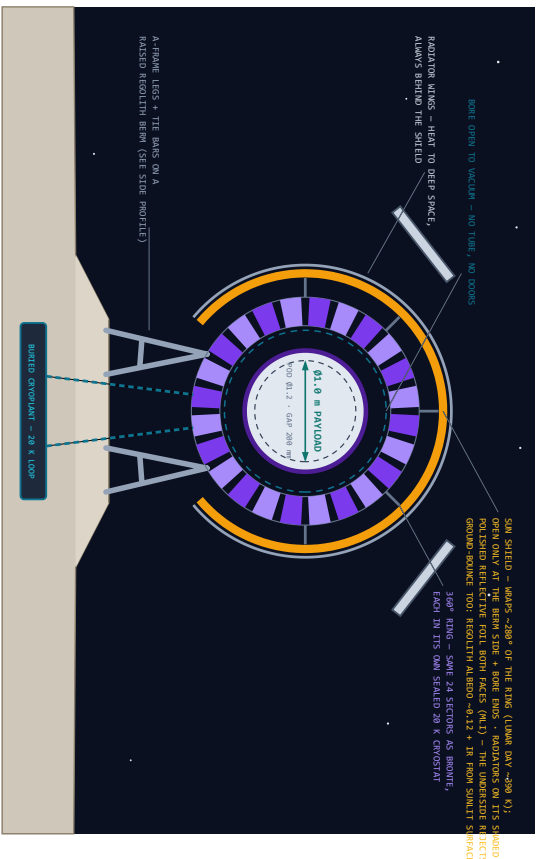
PANEL A – SITE: REAL LRO/LOLA TOPOGRAPHY – SHACKLETON / CONNECTING RIDGE, ~89.7°S



PANEL B – EOS-1 OPEN-TRACK SECTION – NO TUBE: THE MOON IS THE VACUUM VESSEL

EOS-1 SECTION — THE SELENE RING, OPEN TO VACUUM — Ø1.6 m BORE · POD Ø1.2 m · GAP 200 mm

DRAWN TO SCALE (1 m = 110 px) • 24-SECTOR REBCO RING @ 20 K IN SEALED CRYOSTATS • WRAP-AROUND SUN SHIELD COVERS ~280° OF THE RING



POD "EOS-1" - DAWN, SISTER OF SELENE - MISSION ONE: CARGO HOME

ENGINE (REAR) – MIDCOURSE + DROGUE + MAIN CHUTES
EARTH-TARGETING CORRECTIONS
PICA HEAT SHIELD (FRONT) – DOUBLES AS
THE DUST DOME • FLIES SHIELD-FIRST

FULL-HULL REACTION SLEEVE - SAME 360° COUPLING, GAP 280 mm
 Ø1.2 m x 7 m • 500 kg GROSS • REAR ENGINE ΔV 0.77 km/s (Isp 350) • RCS QUADS, NO FINS

CANISTERS FIRST • SATELLITE DISPENSERS LATER • SAME POD, SAME RING

DIRECT SHIELD-FIRST ENTRY → CHUTES → RECOVERY. THE POOL IS THE SHIPPING CRATE.

HARDENED TO 100 g - ARTILLERY ELECTRONICS QUALITY TO 15,000-20,000 g

THE HONEST HARD PARTS (LUNAR EDITION)

1. DUST - ELECTROSTATIC + EJECTA, SHIELD WRAP, PLINTHS, AND THE DOME EXIST BECAUSE OF IT. APOLLO'S #1 MECHANICAL COMPLAINT.

2. THERMAL SINKS - 390 K SUN TO CRYO-COLD SHADOW ON ONE TRACK. ANSWERED BY THE 280° SHIELD + SEALED CRYOSTATS + BURIED LOOP.

3. THE PULSE - 0.39 MM IN 2.45 = 1.17 GM FROM FLAMELETS CHARGED BY ~10 MM OF RIDGE SOLAR. STORAGE MASS RIDES THE FIRST ROCKETS.

5. ICE UNCERTAINTY - SHACKLETON BOUNDED 55-10 wt%, UNPROVEN. THE CATCHER'S MITT (SHIELD + CHUTES), CUSUMAR CATCH COMES LATE

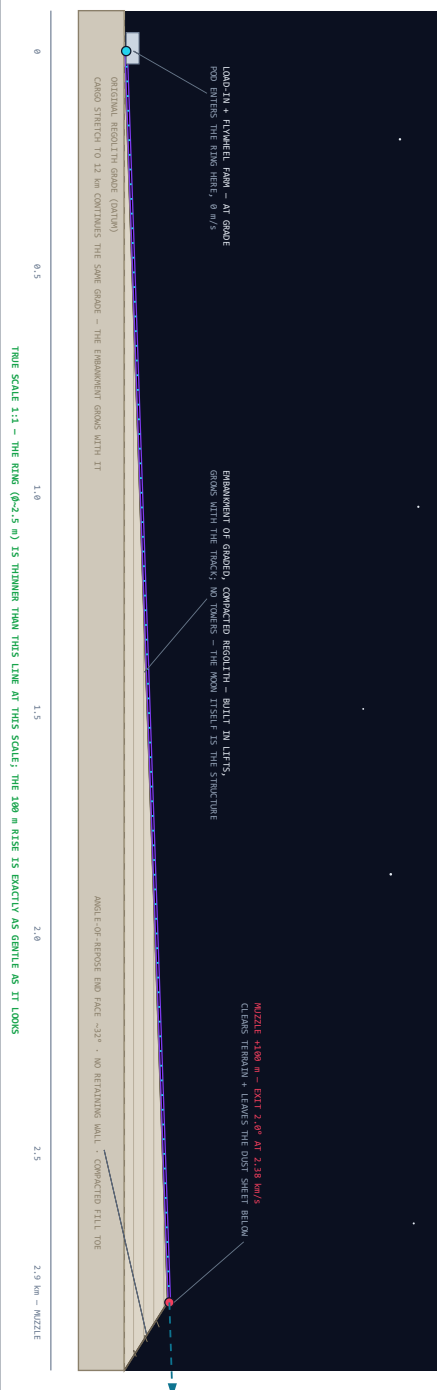
PLAN CLOSES ON REGOLITH/O₂/METAL IN THE LEAN CASE.

MISSING FROM THIS LIST IS THE POINT: NO WINDOW, NO TUBE, NO BOOM, NO LAUNCH HEATING, NO WEATHER, NO RESERVE LAWYERS.

PANEL C – SIDE PROFILE: THE RAISED TRACK ON ITS EMBANKMENT (TRUE SCALE 1:1)

EOS-1 POCKET TRACK — 2.9 km AT A GENTLE 2.0° GRADE, RISING ~100 m ON A BUILT REGOLITH EMBANKMENT

TRUE SCALE 1:1 (0.47 px/mm) • LOAD-IN AT GRADE → MUZZLE +100 m • IN VACUUM THE ANGLE IS FOR TERRAIN + DUST CLEARANCE, NOT PHYSICS



PANEL A2 – WHERE ON THE MOON • WHAT DOES NOT EXIST HERE • TWO MACHINES

WHERE ON THE MOON?



HERE – SOUTH POLE (BOTTOM EDGE, AS SEEN FROM EARTH)

PHOTO: REAL NEARSIDE (FULL MOON) - PANEL A MAP: REAL_LRO/LOLA LASER TOPOGRAPHY

WHAT DOES NOT EXIST ON THIS SHEET

NO VACUUM TUBE – THE TRACK IS ALREADY IN HARD VACUUM

NO PLASMA WINDOW – NO ATMOSPHERE TO KEEP OUT

NO AERO-SHELL, NO NEEDLE NOSE – NOTHING TO CUT THROUGH

NO ISOLATION GATES, NO BLAST SHUTTERS – NO TUBE TO BREACH

NO SONIC BOOM, NO LAUNCH HEATING, NO DRAG – EVER

EVERY HARD SUBSYSTEM ON THE BRONTE SHEET IS DELETED BY ONE SENTENCE: THE MOON HAS NO ATMOSPHERE. WHAT REMAINS IS THE RING, THE POWER, THE COLD – AND THE MOON SUPPLIES THE COLD.

WHAT REPLACES THEM: DUST, 250-300 K THERMAL SWINGS, GW-CLASS PULSE STORAGE, AND CATCHING THE CARGO ON THE OTHER END.

SITE SERVICES: RIDGE SOLAR 80-90% UPTIME • PSR COLD SINK NEXT DOOR • REGOLITH ~250 K AT 2 m DEPTH – NATURAL CRYO BUFFER

MAP DATA: NASA LRO/LOLA POLAR DEM, 20 m/px (LDEM_B755_20M), POLAR STEREOGRAPHIC, RENDERED AS HILLSHADE – A REAL SURVEY, UNLIKE EVERYTHING ELSE ON THIS CONCEPT SHEET.

TWO MACHINES, ONE RIDGE – BUILD SMALL, THEN WIDE

EOS-1 – FIRST BUILD • POD Ø1.2 m • BORE Ø1.6 m • GAP 200 mm

500 kg GROSS • CARGO RETURN TO EARTH IS MISSION ONE

0.39 MWh/SHOT • PEAK 1.17 GW • RECHARGE 2.4 min @ 10 MW

~25 SHOTS/HR → ~300 t/DAY THROWN FROM ONE STARTER TRACK

EOS-2 – SECOND MACHINE • POD Ø2.0 m • BORE Ø2.4 m

~2.5 t GROSS • BIGGER UNIT CARGO + SATELLITE DISPENSERS

1.96 MWh/SHOT • PEAK 5.8 GW • ~5 SHOTS/HR @ 10 MW

SECOND PAD, SAME RIDGE – BUILT AFTER EOS-1 PROVES THE KIT

SAME ~300 t/DAY EITHER WAY – THROUGHPUT IS SET BY THE SOLAR PLANT, NOT THE POD. EOS-2 BUYS SIZE, NOT MASS.

LADDER: 2.9 km @ 100 g → 12 km @ 30 g → ~50 km @ 3 g

ENERGY: 0.41 ORBIT • 0.78 ESCAPE • 1.25 MARS XFER kWh/kg

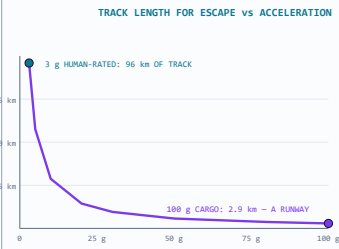
PELLET MODE (SJSU 2023): 25.4 kg EVERY 10 s ON 8.7 MW

FIRST BILL IS A ROCKET BILL: STARTER COILS + PLANT MUST FIT 1-2 HEAVY-LIFT FLIGHTS

LAUNCH AZIMUTH ALONG THE RIDGE; ORBIT PLANE PICKED BY LAUNCH TIME – A POLAR SITE COVERS ALL GEOMETRY MONTHLY

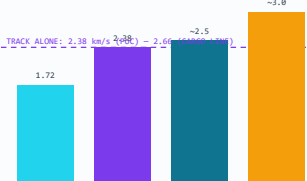
DETAILS – SITE & SYSTEM CHARTS

TRACK LENGTH FOR ESCAPE vs ACCELERATION



$L = v^2/2a$ AT $v = 2.38$ km/s, ACCELERATION TOLERANCE IS THE WHOLE COST MODEL: HARDENED CARGO MAKES THE STARTER TRACK TINY.

WHERE THE TRACK + KICK GETS YOU (Δv FROM SURFACE)



THE 0.77 km/s KICK STAGE TOPS UP THE REST: EVERYTHING IN CISLUNAR SPACE – AND A MARS SHIPMENT – IS IN REACH.

RIDGE ILLUMINATION OVER A YEAR (LITERATURE, SCHEMATIC)

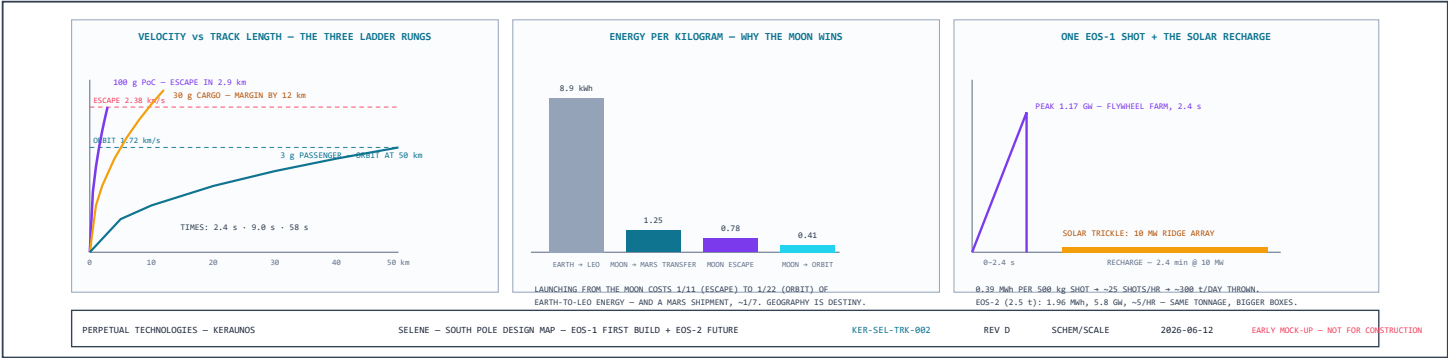


GOLD = SUNLIT (80-90% OF THE YEAR AT THE BEST RIDGE SITES)

DARK SPELLS RUN HOURS TO A FEW EARTH-DAYS (LONGEST KNOWN OUTAGES ~3-5 DAYS) – BRIDGED BY THE SAME FLYWHEEL FARM THAT FIRES THE TRACK; LAUNCHES PAUSE, NOTHING FREEZES.

WHY NO MONTHS-LONG EXTREMES: LUNAR AXIAL TILT IS ONLY 1.5° – THE MOON BARELY HAS SEASONS. THE 1130 °C SWINGS ARE THE EQUATOR'S 14-DAY NIGHT PROBLEM; POLAR RIDGES ARE THE MOST THERMALLY STABLE GROUND ON THE MOON. (TIMELINE SCHEMATIC.)

DETAILS — SELENE ENGINEERING CHARTS



REV D — pod architecture per direction: the PICA heat shield moves to the FRONT, where it pulls double duty as the launch dust dome (one blunt face serves both worlds — dust at launch, plasma at Earth), and a correction engine sits at the REAR for midcourse and Earth-targeting burns; the pod flies shield-first the whole trip. The section is now titled the SELENE ring (parts-shared with BRONTE, but its own machine). The ground geometry is rebuilt — one continuous regolith polygon with a flat-topped berm the legs actually stand on — and a new Panel C shows the BRONTE-style side profile at true 1:1: the 2.9 km track rising ~100 m at 2.0° on a growing regolith embankment built up in compacted lifts and ending in an angle-of-repose (~32°) face rather than a cliff, load-in at grade, muzzle elevated to clear terrain and the dust sheet, with the honest vacuum note that the angle is for clearance, not physics. Prior note: the Moon is now real: Panel A is a hillshade render of NASA LRO/LOLA laser topography (20 m/px polar DEM; projection verified by landing the test markers inside Shackleton itself), and the locator inset uses a real nearside photograph. The sun shield gained its material spec — polished reflective foil on both faces, because the underside sees regolith ground-bounce (albedo ~0.12) and surface IR even though the sun never gets beneath it. A new charts row covers what the old sheet only asserted: track-length-versus-g (why hardened cargo makes the starter track runway-sized), the Δv bars showing track-plus-kick reaching everything cislunar plus a Mars shipment, and the ridge illumination year with its few-day dark spells bridged by the same flywheels that fire the track. Prior note: geometry regenerated clean (24 even sectors, symmetric A-frames with tie bars; no more wonk), the sun shield now wraps ~280° of the ring — a gold thermal shell open only toward the berm and the bore ends, with the radiator wings mounted on its permanently shaded back — and Panel A opens with a "where on the Moon?" locator: this is the very bottom of the Moon, ~89.7°S, the Shackleton–de Gerlache Connecting Ridge. The pod line is now two machines: **EOS-1 first** — Ø1.2 m × 7 m, 500 kg, bore Ø1.6 m at the standard 200 mm gap, whose payload bay takes interchangeable racks (cargo-return canisters first, satellite dispensers later) and whose mission one is shipping lunar material home: escape at 2.38 km/s, kick trim, shield-first reentry, chutes — the pod is the shipping crate, and the heat shield earns its mass only at Earth. **EOS-2 follows** — Ø2.0 m, ~2.5 t, its own pad on the same ridge — and the charts carry the quiet insight: at a fixed 10 MW solar plant both machines throw the same ~300 tonnes a day; the bigger ring buys bigger boxes, not more mass. Throughput belongs to the power plant.